stores the power during expansion stroke and releases the energy during other strokes. It helps crank shaft to rotate continuously. The flywheel of any engine is generally made of cast iron or brass steel. Ring gear is situated at the periphery of the flywheel. This ring gear is in mesh with the pinion gear of the start motor which is used to start the engine.

5.3.16 Silencer

TLVN5J

After the power stroke of the engine the exhaust gases get expanded and expelled through the exhaust manifold. During the exhaust process when the gases are passed through exhaust manifold, due to pressure differential huge noise is produced. This leads to noise pollution. In order to reduce the noise pollution and to operate the engine in smooth condition silencer is used. Silencer acts as resistant for noise and converts the sound energy into heat. Fibre glass is generally used as the insulation material inside the silencer. Chambered, turbo, straight flow are the types of silencer used in automobile.



Figure 5.3.16 Silencer

5.4 FOUR STROKE PETROL ENGINE

Engine running with petrol as a fuel is called as petrol engine. If a power stroke is obtained once in every four stroke of piston (TDC to BDC / BDC to TDC), then the engine is said to be four stroke petrol engine.

Construction

The figure shows the construction of a single cylinder petrol engine. In this the reciprocating motion of a cylinder is converted into rotary motion of crankshaft with the help of connecting rod. One end of connecting rod is connected to piston and another end it is connected to crankshaft. In crankshaft, flywheel is attached at one end and vibration damper, fan belt pulley is attached at the other end. Intake valve, exhaust valve, spark plug is mounted above the top of the cylinder. The fresh air fuel mixture is inducted through intake valve and the burnt gases are sent out through exhaust valve. The opening and closing of valve are made by camshaft. The camshaft and crankshaft are connected by means of timing gear. The four stroke petrol engine is shown in Figure 5.4.

In any internal combustion engine, the following definite sequence of events called strokes namely

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